

1. Title: NetShield AI: Network Anomaly Detection & Threat Monitoring System

2. Objective:

Build an AI-powered network anomaly detection platform that continuously monitors network traffic, identifies suspicious behavior, predicts potential intrusions, and generates real-time threat alerts.

The system should support network traffic monitoring, anomaly detection, intrusion prediction, threat intelligence analysis, and security alert management through a centralized platform.

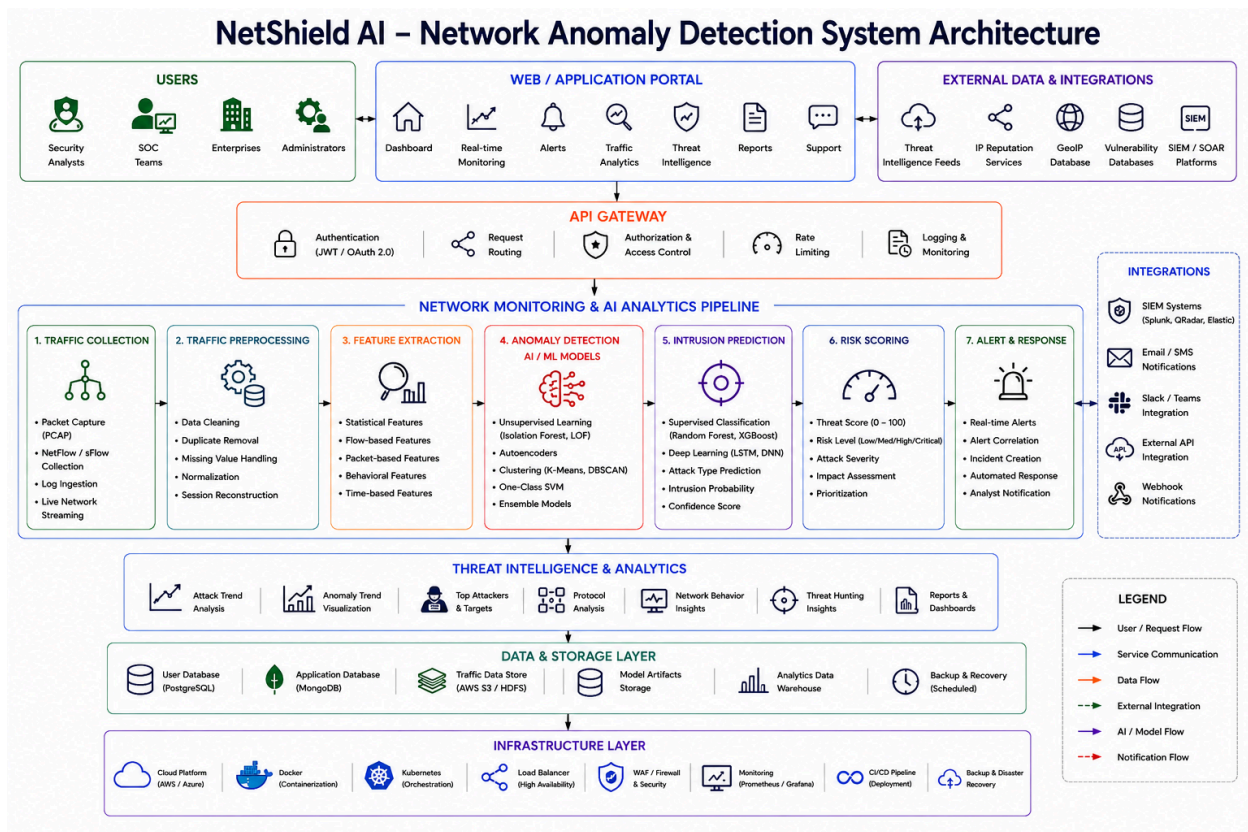
The platform is designed to improve cybersecurity posture by detecting unusual network activities before they escalate into security incidents.

This solution can be used by enterprises, SOC (Security Operations Center) teams, educational institutions, cloud providers, and cybersecurity organizations.

Outcomes:

- Developed and deployed an AI-powered network anomaly detection and threat monitoring platform.
 - Implemented authentication and role-based access control systems.
 - Built network traffic monitoring and packet analysis workflows.
 - Developed anomaly detection and intrusion prediction models using machine learning.
 - Implemented threat alerting and incident management systems.
 - Built analytics dashboards for traffic monitoring and threat intelligence reporting.
 - Developed AI-powered attack classification and risk scoring engines.
 - Deployed the platform using Docker and cloud deployment platforms such as AWS or Azure.
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3. Architecture Diagram:



4. Modules to be Implemented

1. User Management Module

- Security analyst login
- Role-based access control
- Team management
- Audit logging

2. Network Monitoring Module

- Packet collection
- Traffic monitoring

- Protocol analysis
- Traffic visualization

3. Anomaly Detection Module

- Network behavior analysis
- Suspicious activity detection
- Pattern recognition
- Anomaly classification

4. Intrusion Prediction Module

- Threat prediction
- Attack probability analysis
- Risk scoring
- Intrusion forecasting

5. Alert Management Module

- Real-time threat alerts
- Alert prioritization
- Incident management
- Security notifications

6. Analytics Dashboard Module

- Traffic analytics
- Threat intelligence reports
- Security metrics
- Attack trend monitoring

7. AI Detection Module

- Machine learning model training

- Attack classification
 - Behavioral analysis
 - Prediction workflows
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5. Week-wise Module Implementation and High-level Requirements

Milestone 1: Week 1 & 2 — Project Initialization, Design Process & Core Setup

- Define security monitoring objectives and cybersecurity workflows.
- Design system architecture and database schema.
- Create UI wireframes and workflow planning.
- Setup frontend and backend environments.
- Implement authentication and role-based access system.
- Load CICIDS2017 and UNSW-NB15 datasets.
- Build packet monitoring workflows and traffic analytics dashboard.

Outcomes:

- Understand network security monitoring and SOC workflows.
 - Learn system architecture and database design concepts.
 - Build frontend and backend project initialization.
 - Working authentication and network monitoring system.
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Milestone 2: Week 3 & 4 — Anomaly Detection & Intrusion Prediction

- Train anomaly detection models.
- Evaluate model performance and accuracy.
- Generate anomaly detection reports.

- Implement attack prediction workflows.
- Build threat classification modules.
- Generate risk scoring system.

Outcomes:

- Implement anomaly detection and intrusion prediction systems.
 - Build AI-powered threat analysis workflows.
 - Understand machine learning applications in cybersecurity.
 - Generate real-time threat detection insights.
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Milestone 3: Week 5 & 6 — Alert Management & Security Analytics

- Implement threat alerting workflows.
- Build notification and incident management systems.
- Generate threat intelligence reports.
- Build attack visualization dashboards.
- Implement monitoring and reporting workflows.
- Develop security analytics dashboards.

Outcomes:

- Build threat alerting and analytics systems.
 - Implement monitoring and incident response workflows.
 - Understand cybersecurity operations and threat intelligence concepts.
 - Complete end-to-end network security monitoring workflows.
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Milestone 4: Week 7 & 8 — Testing, Deployment & Documentation

- Validate AI model performance and prediction accuracy.
- Optimize traffic processing and dashboard responsiveness.

- Deploy platform using Docker and cloud environments.
- Prepare final project documentation and presentation.
- Demonstrate the complete NetShield AI platform.

Outcomes:

- Gain deployment and testing experience.
 - Improve detection accuracy and platform usability.
 - Complete live deployment and final demonstration.
 - Prepare professional project documentation and presentation.
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6. Evaluation Criteria

Milestone 1 (Week 2):

- Project initialization and architecture setup completed.
 - Authentication and network monitoring workflows implemented.
 - Traffic analytics dashboard functional.
 - Dataset integration and preprocessing completed.
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Milestone 2 (Week 4):

- Anomaly detection and intrusion prediction workflows implemented.
 - Threat classification and risk scoring functional.
 - Detection reports generated successfully.
 - AI models integrated with monitoring workflows.
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Milestone 3 (Week 6):

- Alert management and analytics dashboard implemented.
- Threat intelligence reporting functional.
- Attack visualization dashboards generated.

- Monitoring and notification workflows integrated.
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Milestone 4 (Week 8):

- Fully deployed frontend and backend.
 - Model testing and validation completed.
 - Documentation and presentation prepared.
 - Successful end-to-end platform demonstration completed.
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7. Tools & Tech Stack

Programming Language

- Backend: Python (FastAPI)
 - Frontend: React.js / Next.js
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Database

- PostgreSQL
 - MongoDB
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AI & Machine Learning

- Scikit-learn
 - TensorFlow
 - XGBoost
 - Pandas
 - NumPy
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Network Monitoring

- Wireshark
- Zeek

- Packet Capture Tools
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Security Integrations

- SIEM Platforms
 - Threat Intelligence APIs
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Cloud & DevOps

- Docker
 - AWS / Azure
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Libraries & Frameworks

- FastAPI
 - React.js
 - Next.js
 - Tailwind CSS
 - JWT Authentication
 - TensorFlow
 - XGBoost
 - Chart.js
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Dev & Deployment Tools

- IDE: VS Code
 - Version Control: Git + GitHub
 - Docker & Docker Compose
 - Deployment: AWS / Azure
 - API Testing: Postman
 - Monitoring: Optional Logging & Monitoring Tools
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8. Performance Metrics

AI Model Performance

- Detection accuracy
 - Precision
 - Recall
 - F1-score
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Security Performance

- Threat detection rate
 - False positive rate
 - Alert generation latency
 - Attack classification accuracy
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System Performance

- Traffic processing speed
 - Dashboard response time
 - Concurrent monitoring capability
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9. Example Quantitative Goals

Anomaly Detection

- Achieve accurate detection of unusual network behavior and suspicious activities.
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Intrusion Prediction

- Predict potential cyberattacks and generate reliable risk assessments.
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Threat Monitoring

- Provide real-time threat intelligence and alerting for security teams.
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Platform Performance

- Support large-scale network monitoring and threat analysis with stable system performance.